Trigonometry: Trigonometric Identities in Radians

Questions 1–28

Question 1. It is given that $\sin^2 x = \frac{4}{5}$ and x is acute. Find the exact values of:

a) $\cos x$

b) $\tan x$

Question 2. It is given that $\cos^2 x = \frac{8}{9}$ and x is acute. Find the exact values of:

a) $\sin x$

b) $\tan x$

Question 3. It is given that $\tan^2 x = \frac{25}{6}$ and x is acute. Find the exact values of:

a) $\sin x$

b) $\cos x$

Question 4. It is given that $\sin^2 x = \frac{5}{12}$ and x is obtuse. Find the exact values of:

a) $\cos x$

b) $\tan x$

Question 5. It is given that $\cos^2 x = \frac{12}{13}$ and x is obtuse. Find the exact values of:

a) $\sin x$

b) $\tan x$

Question 6. It is given that $\tan^2 x = \frac{30}{7}$ and x is obtuse. Find the exact values of:

a) $\sin x$

b) $\cos x$

Question 7. It is given that $\sin^2 x = \frac{3}{16}$ and $\pi < x < \frac{3\pi}{2}$. Find the exact values of:

a) $\cos x$

b) $\tan x$

Question 8. It is given that $\cos^2 x = \frac{11}{12}$ and $\pi < x < \frac{3\pi}{2}$. Find the exact values of:

a) $\sin x$

b) $\tan x$

Question 9. It is given that $\tan^2 x = \frac{18}{11}$ and $\pi < x < \frac{3\pi}{2}$. Find the exact values of:

a) $\sin x$

b) $\cos x$

Question 10. It is given that $\sin^2 x = \frac{24}{25}$ and $\frac{3\pi}{2} < x < 2\pi$. Find the exact values of:

a) $\cos x$

b) $\tan x$

Question 11. It is given that $\cos^2 x = \frac{24}{49}$ and $\frac{3\pi}{2} < x < 2\pi$. Find the exact values of:

a) $\sin x$

b) $\tan x$

Question 12. It is given that $\tan^2 x = \frac{60}{17}$ and $\frac{3\pi}{2} < x < 2\pi$. Find the exact values of:

a) $\sin x$

b) $\cos x$

Question 13. It is given that $\cos\frac{\pi}{10} = \sqrt{\frac{5}{8} + \frac{\sqrt{5}}{8}}$

a) Find the exact value of $\sin\frac{\pi}{10}$

b) Using your answer to a), show that $\tan\frac{\pi}{10} = \sqrt{1 - \frac{2}{\sqrt{5}}}$

Question 14. Serkan is attempting to solve the equation $3\sin x = \tan x$ for $0° \leq x < 360°$. His attempt is shown below:

$$\begin{aligned} 3\sin x &= \tan x \\ \Rightarrow 3\sin x &= \frac{\sin x}{\cos x} \\ \Rightarrow 3\sin x\cos x &= \sin x \\ \Rightarrow 3\cos x &= 1 \\ \Rightarrow \cos x &= \frac{1}{3} \\ \Rightarrow x &= 1.23 \text{ or } 5.05 \text{ to 3sf} \end{aligned}$$

Explain the two errors that Serkan has made.

Question 15. Solve the following equations:

a) $\sin\theta = \cos\theta,\ 0 \leq \theta < 4\pi$

b) $20\sin\theta + 3\cos\theta = 0,\ 0 \leq \theta < 2\pi$

c) $11\cos\theta - 4\sin\theta = 0,\ 0 \leq \theta < 2\pi$

Question 16. Solve the following equations:

a) $\sin 2\theta = \cos 2\theta,\ 0 \leq \theta < 3\pi$

b) $7\sin 4\theta = 2\cos 4\theta,\ 0 \leq \theta < \frac{\pi}{2}$

c) $10\sin\frac{1}{2}\theta + 3\cos\frac{1}{2}\theta = 0,\ 0 \leq \theta < 5\pi$

d) $19\cos\frac{1}{4}\theta - 16\sin\frac{1}{4}\theta = 0,\ 0 \leq \theta < 20\pi$

Question 17. Solve the following equations:

a) $6\tan\theta + \cos\theta = 0,\ 0 \leq \theta < 2\pi$

b) $\sin\theta\tan\theta + 16 = 0,\ 0 \leq \theta < 2\pi$

c) $12\tan\frac{1}{2}\theta + \cos\frac{1}{2}\theta = 0,\ 0 \leq \theta < 8\pi$

Question 18. Solve the following equations:

a) $3\cos^2\theta = 8\sin\theta,\ 0 \leq \theta < \pi$

b) $9\sin^2\theta = 80\cos\theta,\ 0 \leq \theta < 3\pi$

c) $12\cos^2\theta = 143\sin\theta,\ 0 \leq \theta < 2\pi$

d) $50\sin^2\theta = 2499\cos\theta,\ -\pi \leq \theta < \pi$

Question 19. Solve the following equations:

a) $4\cos^2\theta + 9\sin\theta - 6 = 0,\ 0 \leq \theta < 2\pi$

b) $9\sin^2\theta + 3\cos\theta - 7 = 0,\ 0 \leq \theta < 2\pi$

c) $15\cos^2\theta - 31\sin\theta - 25 = 0,\ 0 \leq \theta < 2\pi$

d) $27\sin^2\theta + 12\cos\theta - 7 = 0,\ 0 \leq \theta < 2\pi$

Question 20. Solve the following equations:

a) $8\sin^2\theta - 26\sin\theta - 16\cos^2\theta + 21 = 0,\ 0 \leq \theta < 2\pi$

b) $13\cos^2\theta + 13\cos\theta - 22\sin^2\theta + 10 = 0,\ 0 \leq \theta < 2\pi$

c) $20\sin^2\theta - 4\cos^2\theta + 38\sin\theta - 51 = 0,\ 0 \leq \theta < 2\pi$

d) $50\sin^2\theta + 28 = 30\cos^2\theta + 31\cos\theta,\ 0 \leq \theta < 2\pi$

Question 21.

a) Solve the equation $20\sin^2 x - 14\sin x - 1 = 4\cos^2 x$ for $-2\pi \leq x < 2\pi$

b) Hence find the smallest positive solution to the equation $20\sin^2(3\theta + \frac{\pi}{3}) - 14\sin(3\theta + \frac{\pi}{3}) - 1 = 4\cos^2(3\theta + \frac{\pi}{3})$

Question 22. Solve the following equations:

a) $36\sin\theta + \frac{10}{\cos\theta} = 24\tan\theta + 15,\ 0 \leq \theta < 2\pi$

b) $3\sin\theta\tan\theta = 21\cos\theta + 14,\ 0 \leq \theta < 2\pi$

c) $\frac{6\cos\theta}{\tan\theta} = 29\sin\theta - 1,\ 0 \leq \theta < 2\pi$

d) $10\sin\theta\cos\theta - 2\cos^2\theta + 5\cos\theta = 2\sin\theta(2 + \sin\theta),\ 0 \leq \theta < 2\pi$

Question 23. Solve the following equations:

a) $3\sin\theta\tan\theta = 2\sin\theta + \cos\theta,\ 0 \leq \theta < 2\pi$

b) $8\sin\theta\tan\theta + 39\sin\theta = 5\cos\theta,\ 0 \leq \theta < 2\pi$

c) $74\cos\theta\tan\theta = 45\sin\theta\tan\theta + 24\cos\theta,\ 0 \leq \theta < 2\pi$

d) $40\sin\theta\tan\theta = 15\cos\theta + 38\cos\theta\tan\theta,\ 0 \leq \theta < 2\pi$

Question 24.

a) Solve the equation $18\sin^2 x + 3\cos x = 8,\ 0 \leq x < 4\pi$

b) Hence solve the equation $18\sin^2 \frac{1}{3}\theta + 3\cos\frac{1}{3}\theta = 8,\ 0 \leq \theta < 12\pi$

Question 25.

a) Show that $\dfrac{2\sin x - 5\cos^2 x - 2}{5\sin x + 7} \equiv \sin x - 1$

b) Hence solve the equation $\dfrac{2\sin(\theta + \frac{\pi}{3}) - 5\cos^2(\theta + \frac{\pi}{3}) - 2}{5\sin(\theta + \frac{\pi}{3}) + 7} = 10\cos(\theta + \frac{\pi}{3}) - 1$ for $0 \leq \theta < 2\pi$

Question 26.

a) Show that $\dfrac{25 - 40\cos^2 x - 38\sin x}{4\sin x - 5} \equiv 3 + 10\sin x$

b) Hence solve the equation $\dfrac{25 - 40\cos^2 2\theta - 38\sin 2\theta}{4\sin 2\theta - 5} = 3 - 8\cos 2\theta$ for $0 \leq \theta < \pi$

Question 27. Solve the equation $\sqrt{18\sin^2\theta - 21\sin\theta\cos\theta} = 2\cos\theta$ for $0 \leq \theta < 2\pi$.

Question 28. Prove the following trigonometric identities:

a) $\frac{\cos^2\theta - \cos^4\theta}{2\cos^2\theta - 1} \equiv \frac{\sin\theta\cos\theta}{\frac{1}{\tan\theta} - \tan\theta}$

b) $\frac{\sin^4\theta - \sin^2\theta}{\sin^3\theta + 2\sin^2\theta - \sin\theta - 1} \equiv \frac{\sin\theta\cos\theta}{\frac{1}{\tan\theta} - \tan\theta + \cos\theta}$